is coupled with the other brake system that enhances and converts the small power of the driver in to multiple times of the power for braking the vehicle. This type of brake system is called as the servo brake. Generally, by using less amount of force getting more amount of power is called as servo action. In this way, using additional power from the high pressure air or from vacuum the hydraulic brakes are operated. This type of brake is called as servo brake or energized brake.

Air servo brake or air assisted hydraulic brake

In this type of brakes for increasing the force on a piston of the master cylinder, compressed air is used. In this type of braking system half of the pressure applied to the pedal from the simple hydraulic braking is sufficient to operate the system. Moreover, even in case of any leakage in the compressed air path, the brakes can be applied by the servo braking system. The schematic of this brake is shown in Figure 6.4.3.1.

For operating the brake system there are two air compressors and control valves used in this system. Apart from these, connecting pipes are present to connect the servo unit with the master cylinder to combine the air pressure with hydro power. When driver presses the brake pedal, the piston lever inside the servo unit is moved forward. At that time the inlet valve in the master cylinder is opened and the compressed air is allowed to enter into the cylinder for increasing the pressure. Hence the brake fluid is sent out with high pressure to the wheel cylinder and operates the brakes. When the driver releases the brake pedal, the outlet valve is opened and the compressed air is sent out through this valve. Hence the brake shoes are released from the brake drum and the drum is allowed to rotate which rotates the wheels.

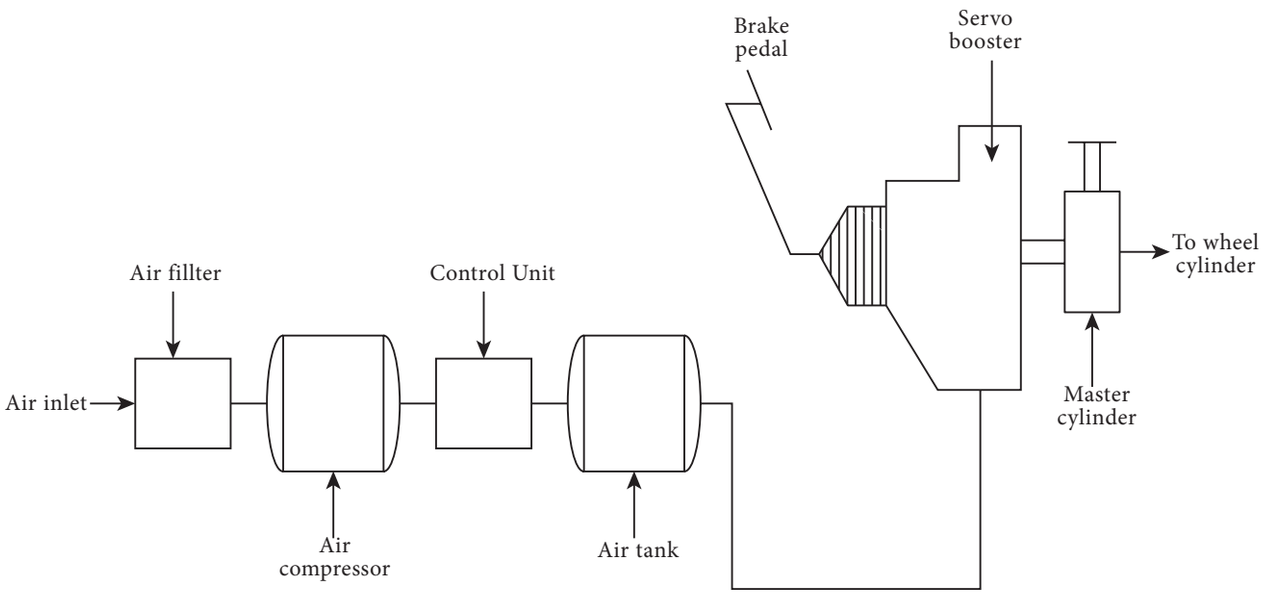


Figure 6.4.3.1 Air servo brake or air assisted hydraulic brake

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